

Elevating User Privileges



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Overview



Becoming the Boss

- Privilege escalation
 - su
 - sudo
 - environment variables and sudo
 - using PolKit (PolicyKit)





UID 0, usually root, manages your Linux System.
Using **su**, **sudo**, **pkexec** we can elevate to UID 0

```
$ su
```

```
$ su -
```

```
$ su -l
```

```
$ su - bob
```

su – Substitute User

Using the command `su` we can change to the root account, or any account that we have the password for. If the root account does not have a password, then this is not an option. We can just change to the account, or we can execute a full login

Having all administration team
all knowing the root password
is not the best of ideas!



Demo



Let's begin by using the su command:

- Login or non-login shell
- Testing user accounts



```
$ sudo less /etc/sudoers  
$ sudo ls /etc/sudoers.d/
```

Sudo

Using sudo, we can delegate administrative tasks without the need to divulge the root password or give access to all commands

```
$ sudo visudo
```

```
$ sudo visudo -f /etc/sudoers.d/bob
```

Editing the Sudoers Files

Using the command `visudo` to make changes to the configuration will enforce a syntax check when the file is saved. A misconfigured sudo entry will disable sudo access to your system.

Demo



Creating Sudoers Entries:

- syntax checking
- delegating tasks to users



```
$ sudo visudo -f /etc/sudoers.d/defaults
Defaults env_keep += "EDITOR"

$ export EDITOR=vim

$ sudo visudo
```

Using Another Editor

In Debian based systems the default editor will be nano, other distributions may use vim. You can set the EDITOR variable but this needs to be allowed to pass through to sudo.

Demo



Using a different editor

- `sudo -i`
- using environment variables with `sudo`



```
$ pkaction
$ sudo cat /etc/polkit1/localauthority.conf.d/51-ubuntu-admin.conf
$ echo $$

$ pktttyagent -p 5296

$ pkexec cat /etc/shadow

$ enter password
```

Using Polkit (Formerly PolicyKit)

Designed more for desktop systems we have PolKit. A separate authenticator program is required to authenticate users. Make sure your user has a password and you belong to the sudo group. We require the authenticator to monitor the process in another window

Demo



Using a default PolKit rules

- use pktttyagent and pkexec
- recover from failed sudo system



Summary



In this module we have introduced privilege escalation in Linux

- su
- su -
- su -l
- sudo
- visudo
- visudo -f
- visudo -c
- Defaults env_keep += "EDITOR"
- pkexec / pktttyagent



Congratulations:
This is the end of the course

